

43 The Immune System

KEY CONCEPTS

- 43.1** In innate immunity, recognition and response rely on traits common to groups of pathogens *p. 953*
- 43.2** In adaptive immunity, receptors provide pathogen-specific recognition *p. 957*
- 43.3** Adaptive immunity defends against infection of body fluids and body cells *p. 963*
- 43.4** Disruptions in immune system function can elicit or exacerbate disease *p. 970*

Study Tip

Make a table: As you study innate and adaptive immunity, compare and contrast these two forms of immune defense by continuing to fill in this table.

Defense mechanism	Key role in innate defense?	Key role in adaptive defense?	Effect on pathogen	Example of cell involved
Barrier	Yes	No	Entry into host tissues prevented	Epithelial cell
Phagocytosis				
Secreted molecules				
Destruction of infected host cells				

Go to Mastering Biology

For Students (in eText and Study Area)

- Get Ready for Chapter 43
- BioFlix® Animation: Activation of a Helper T cell
- BioFlix® Animation: Adaptive Defenses

For Instructors to Assign (in Item Library)

- Activity: The Inflammatory Response
- Tutorial: Acquired Immunity

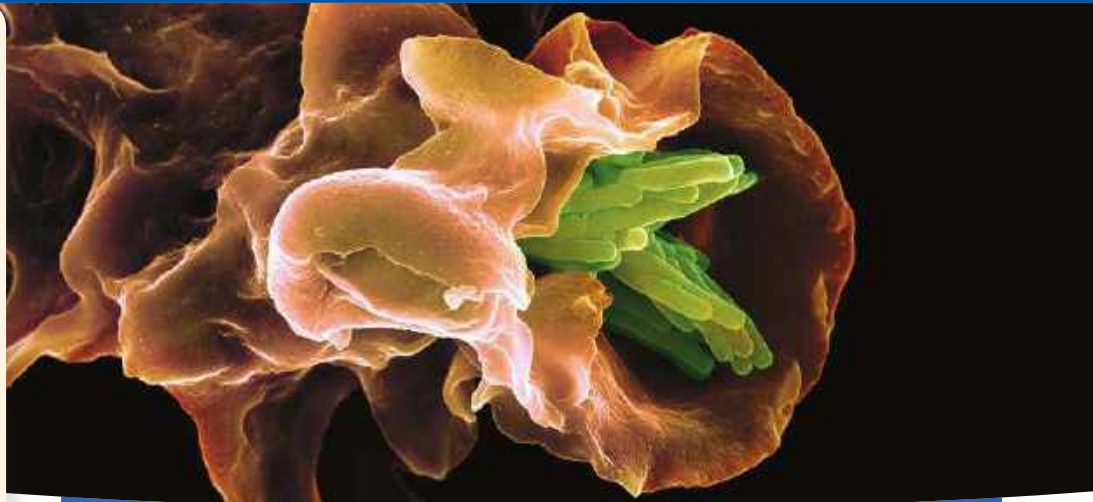


Figure 43.1 Dedicated immune cells in most animals specifically interact with and attack pathogens, disease-causing agents such as bacteria or viruses. Here, an immune cell called a macrophage (brown) is engulfing bacteria (green).

How do immune cells recognize pathogens?

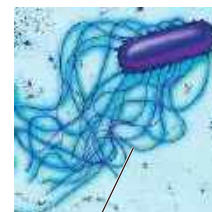
Innate recognition Each of a small set of receptors recognizes a molecule absent from animals, but common to a type of pathogen, as shown by these examples.

Pathogen

Virus

Bacterium

Fungus



Example of recognized molecule

dsRNA (nucleic acid in genome)

Flagellin (protein in flagella)

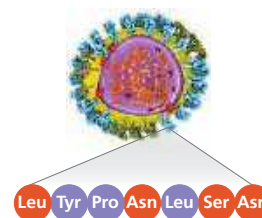
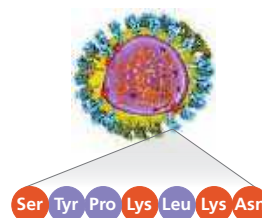
Mannan (oligosaccharide in cell wall)

Adaptive recognition Each of a vast number of receptors is specific for a particular part of a protein in one pathogen, such as a surface protein of the influenza (flu) virus. In this example, a different adaptive immune receptor would recognize each of the flu strains based on the proteins' amino acid sequences, which differ at the positions colored orange.

Influenza virus strain 1

Influenza virus strain 2

Influenza virus strain 3



Recognition by either type of immunity triggers a **response** that can eliminate or inactivate the pathogen, protecting the infected animal.